Major Project

Branch: CSE-Data Science

Department: Data Engineering

Semester: VII **Section: A** **Regulation: R23** **Course Code: R23MSCSP003**

Project Title: LifeSphere AI: A Unified Lifelong Digital Healthcare Ecosystem Using Explainable Artificial Intelligence

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Abstract

Today's healthcare records are often scattered across hospitals, laboratories, and pharmacies, making it difficult for doctors to access a patient's complete medical history. This project proposes LifeSphere AI, a centralized and secure healthcare ecosystem that stores medical records, prescriptions, laboratory reports, and other health information in one place. It allows patients to manage their lifelong health records and provides authorized doctors with quick access to relevant information. The system also uses Machine Learning and Explainable AI to analyze clinical data, assess health risks, identify trends, and provide understandable AI-based insights. Features such as medical timeline, health trends, AI health summary, consent-based access, and audit logging improve transparency and security. By reducing repeated tests, improving information accessibility, and supporting data-driven decisions, LifeSphere AI aims to make healthcare management more efficient and patient-centered.

Introduction

The healthcare sector is rapidly growing, but managing patient medical records still remains a major challenge. In many hospitals and clinics, patient data such as prescriptions, lab reports, and scan results are stored in different places or maintained as physical documents. This makes it difficult for doctors to access complete medical information at the right time. As a result, treatment decisions may be delayed or may not be fully accurate. Patients are often required to carry their reports during every visit, which can be inconvenient. These documents can also be easily lost, damaged, or forgotten. In emergency situations, the absence of medical history can create serious risks for patients. It also leads to repeated medical tests, increasing both time and cost. To solve these issues, there is a need for a centralized system that stores all medical records in one place. A Medical Record Management System helps in organizing and managing patient data efficiently. It allows doctors to quickly access important information and provide better treatment. The system also reduces the chances of data loss and duplication. With secure login and role-based access, it ensures that only authorized users can view or update records. It supports digital storage of all medical data, making the system more reliable and easy to use.

Literature Review/ Background Study

Author & Year	Method / Technology	Key Finding	Limitation / Research Gap
Mohanty et al., 2022 *(Patterns Journal)*	Machine Learning for 30-day Readmission Prediction using EHR with SHAP Explainability	• Evaluated five ML approaches for predicting 30-day unplanned readmission for elderly patients (age ≥ 50). • CatBoost outperformed other models with a mean AUROC of 0.79. • Used a comprehensive set of variables including frailty, comorbidities, high-risk medications, demographics, hospital and insurance utilization. • Provided model interpretability using SHAP.	• Mainly focused on frail/elderly patients and readmission risk. • Performance may vary across institutions and patient populations. • Requires high-quality and standardized EHR data. • Limited integration into real-world clinical workflows.
Choi et al., 2016 *(NeurIPS)*	RETAIN – Reverse Time Attention Mechanism (Interpretable Neural Network for EHR)	• Proposed a two-level attention model that identifies important past visits and significant clinical variables. • Achieved high predictive accuracy with easy interpretability comparable to RNN. • Tested on a large EHR dataset with 14 million visits from 263K patients over 8 years. • Mimics physician behavior by attending past visits in reverse time order.	• Focuses on specific predictive tasks (e.g., diagnosis prediction). • Not a complete healthcare management system. • Requires large sequential EHR data. • Limited to model-level interpretability, not full patient-centric ecosystem.
Rajkomar et al., 2018 *(npj Digital Medicine)*	Deep Learning with Electronic Health Records (EHR) using FHIR Format	• Proposed a deep learning approach using the entire raw EHR data based on the FHIR format. • Achieved high predictive accuracy for multiple clinical outcomes (e.g., in-hospital mortality, AUROC up to 0.93). • Demonstrated scalability across multiple hospitals without site-specific data harmonization.	• Requires large-scale data and computational resources. • Interpretability of deep learning models remains a challenge. • Integration into real-world clinical workflows is still limited. • Focuses on prediction tasks rather than a unified patient health-record platform.

Research Gap: While existing studies demonstrate the potential of machine learning and deep learning with EHRs, provide interpretability techniques, and achieve good predictive performance, there is a need for a unified, patient-centric system that integrates lifelong health-record management with ML-based risk prediction and explainable AI to deliver comprehensive and reliable healthcare services.

Problem Statement

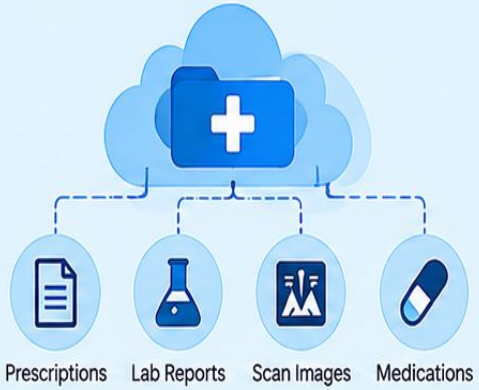
In today's healthcare system, medical records such as prescriptions, laboratory reports, and diagnostic images are stored separately in different hospitals, laboratories, and pharmacies. Because of this, doctors are unable to access a patient's complete medical history at the time of treatment. This leads to repeated tests, delays in diagnosis, and difficulty in understanding previous treatments. Patients often need to carry multiple reports, which can be easily lost or damaged. In emergency situations, missing medical records can affect timely and accurate treatment. Overall, this scattered storage of medical data makes healthcare less efficient.

There is no proper system that collects and stores all medical records like X-rays, scan reports, blood test results, and prescriptions in one place. As a result, tracking a patient's health history over time becomes difficult for doctors. Important past medical information may be ignored or overlooked during treatment. This also makes it harder to understand disease progression and provide better care. A centralized system that stores only medical records can help doctors quickly access required information. Such a system can improve treatment speed, accuracy, and overall patient care.

Project Objectives

01 Centralize

Develop a centralized system to store all patient medical records such as prescriptions, lab reports, scan images, and medical history in one secure



02 Enable Access

Provide easy and quick access to complete medical history for doctors, patients, and authorized healthcare staff, anytime and anywhere.



03 Improve Care

Reduce repetition of medical tests, improve diagnosis accuracy, and support better treatment decisions by utilizing complete and organized patient data.



04 Ensure Security

Ensure secure storage and privacy of medical data using authentication, OTP verification, and role-based access, while providing a user-friendly and reliable platform for all users.



Project Scope

The scope of this project focuses on developing a centralized system for managing patient medical records digitally. It includes storing data such as prescriptions, lab reports, scan images, and medical history in one platform. The system allows patients to access and manage their health records easily. Doctors can view complete patient history for better diagnosis and treatment. Hospitals, laboratories, and pharmacies can upload and update medical data in real time. The system supports role-based access for patients, doctors, and healthcare staff. Secure authentication with OTP verification ensures data privacy and protection. It reduces the need for physical documents and minimizes data loss. The system helps avoid repeated medical tests by providing complete information. It enables quick access to records during emergency situations. The platform is designed to be user-friendly and accessible anytime and anywhere. It improves communication between different healthcare providers. The system maintains continuous patient health tracking over time. It can be extended with AI features for data analysis and prediction. Overall, the project aims to improve healthcare efficiency and patient care.

Data Gathered/ Data Used

1. Synthea – Synthetic Patient Records

- **Source:** MITRE Synthea
- **Records:** 100-patient sample / synthetic patient records
- **Data:** Demographics, encounters, conditions, medications, procedures, allergies, immunizations and observations
- **Format:** CSV / FHIR
- **Purpose:** Building the unified lifelong health record

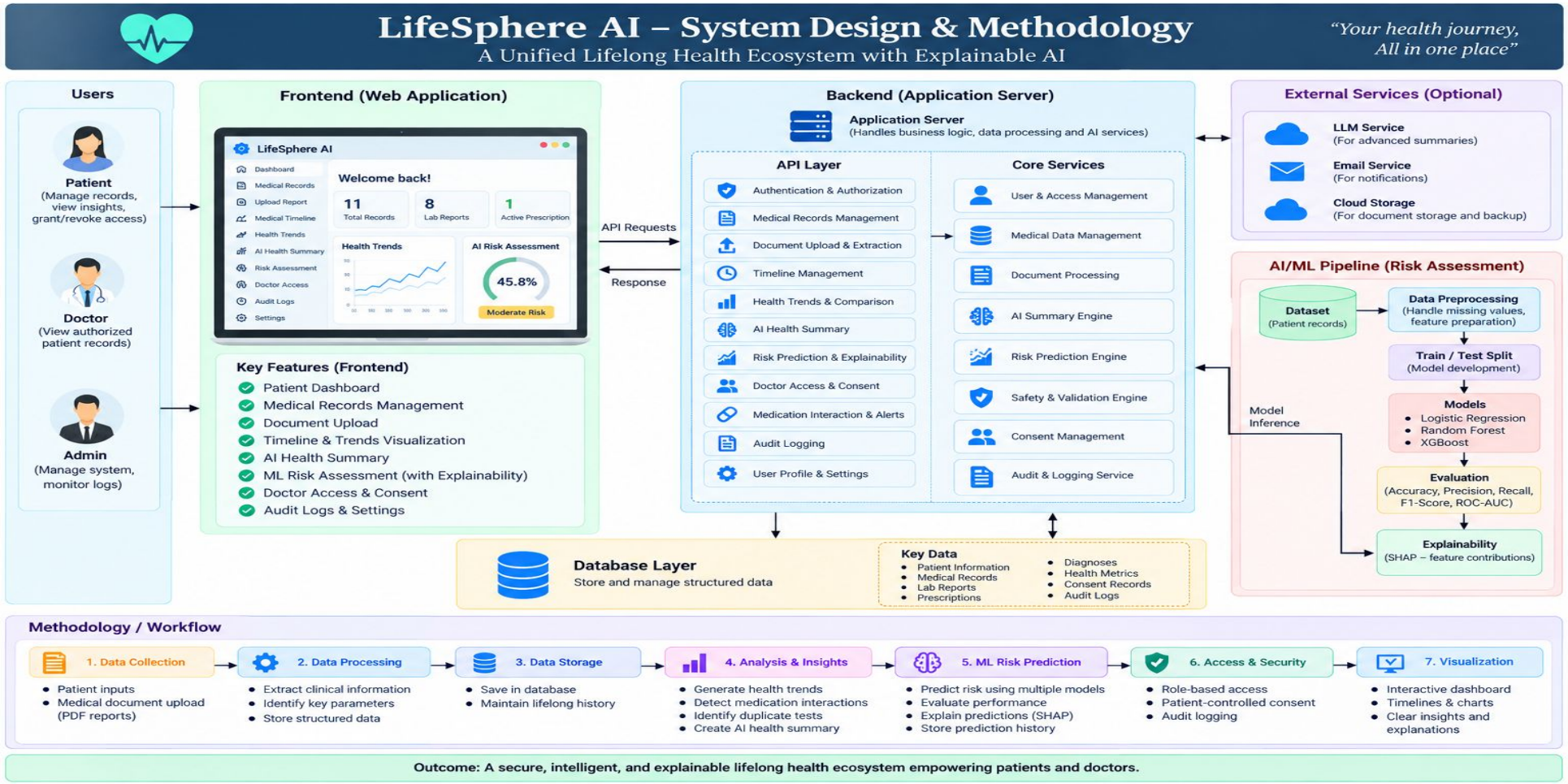
2. Diabetes 130-US Hospitals Dataset

- **Source:** UCI Machine Learning Repository
- **Records:** 101,766 | **Features:** 47
- **Input:** Demographics, admission details, diagnoses, laboratory tests, medications, hospital stay, previous visits, etc.
- **Target:** Early hospital readmission within 30 days
- **Format:** CSV
- **Purpose:** ML-based health/readmission analysis

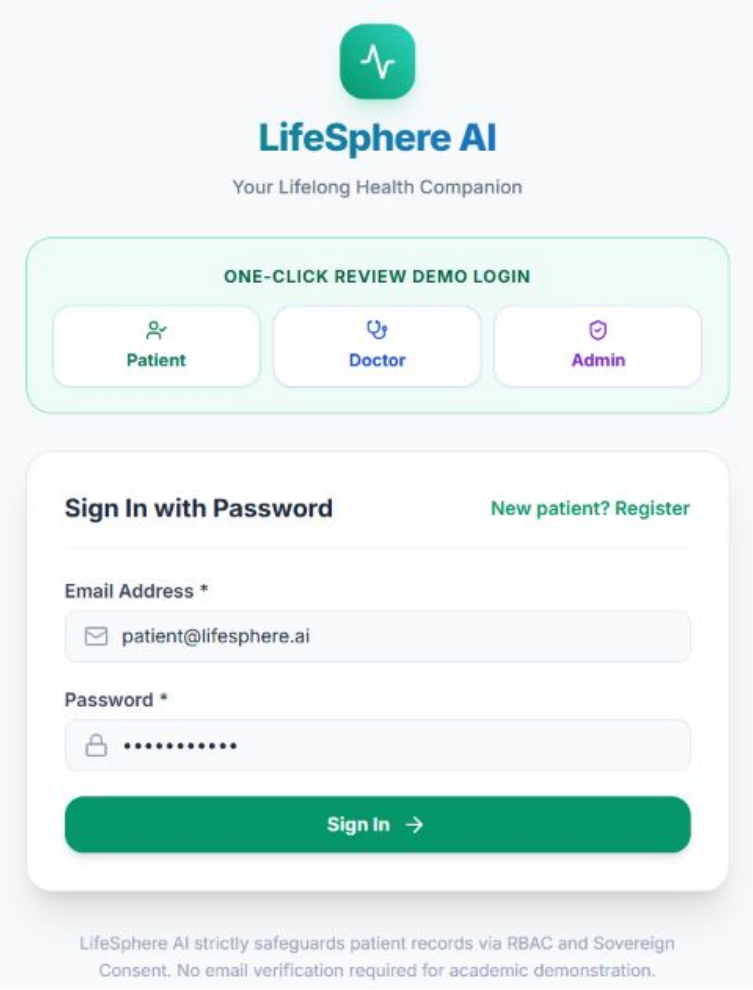
Preprocessing: Missing-value handling → Data cleaning → Feature selection → Data preparation

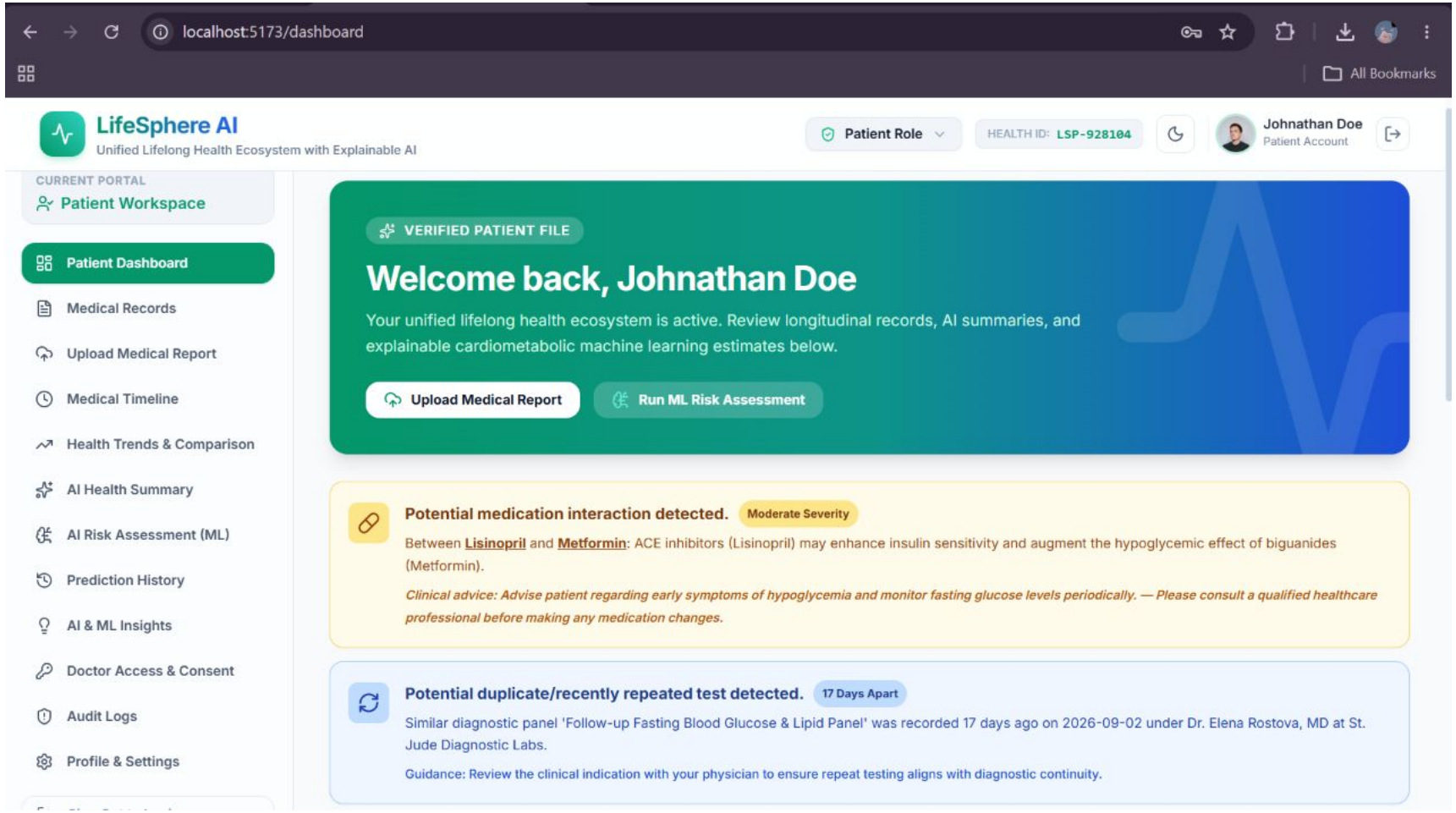
Dataset	Records	Data / Features	Target / Purpose
Synthea	100 synthetic patients	Demographics, encounters, conditions, medications	Lifelong health record
UCI Diabetes	101,766	47 clinical features	30-day readmission

Methodology / System Design

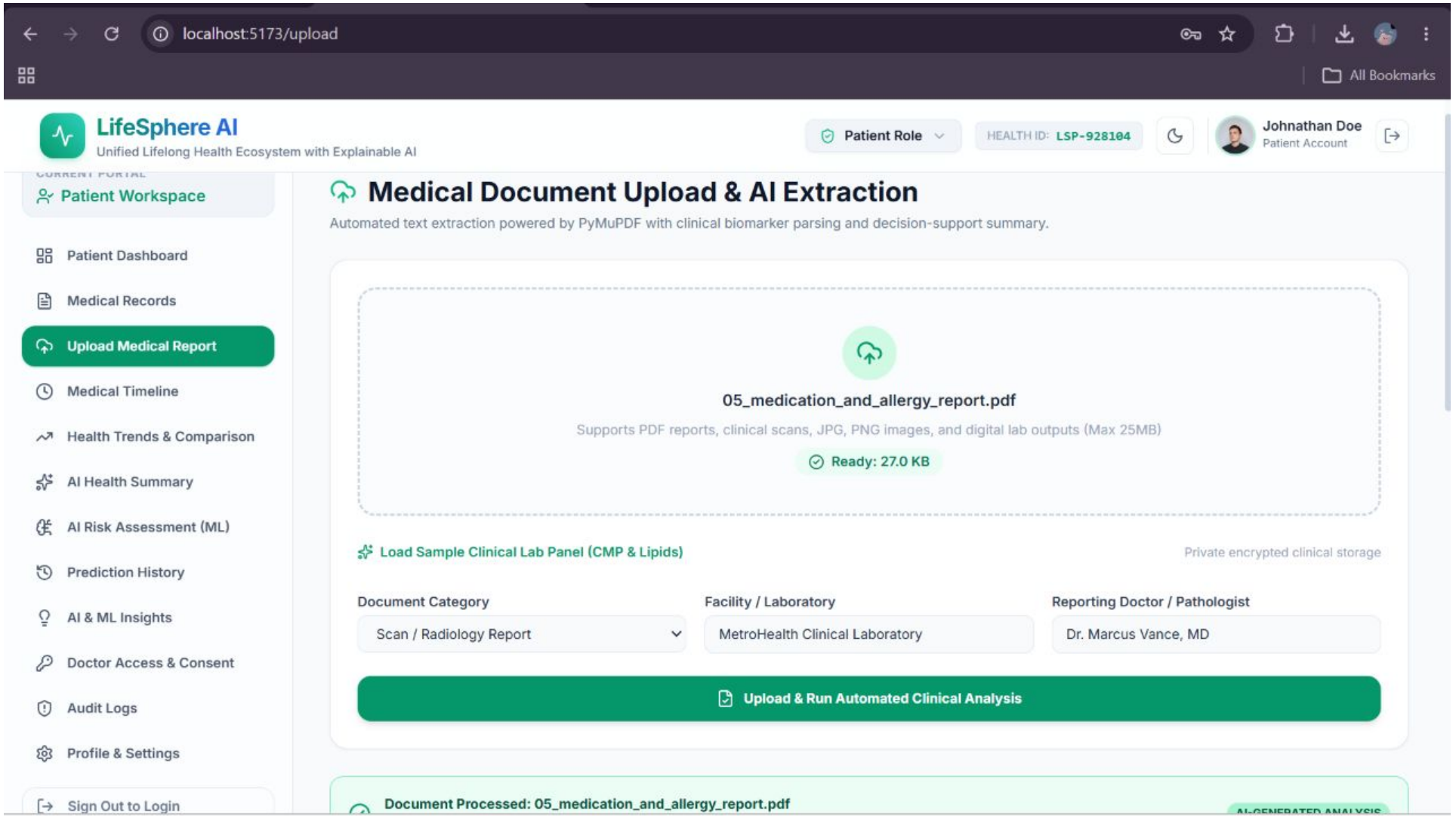


Results (Full) / Screenshots





Results (Full) / Screenshots



Implementation / Modules

CODE SNIPPET:

```
# LifeSphere AI – Readmission Risk Prediction

from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline

model = Pipeline([
    ("preprocessing", preprocessor),
    ("classifier", LogisticRegression(
        class_weight="balanced",
        solver="lbfgs",
        C=0.1
    ))
])

# Train the model
model.fit(X_train, y_train)

# Predict 30-day readmission probability
probability = model.predict_proba(X_test)[:, 1]
```

Impact Assessment

Technical Impact

Digital Records & Accuracy
Improves accuracy by reducing manual errors and storing records digitally.

Fast Patient Data Access
Provides fast access to patient data, helping doctors make quick decisions.

Automation & Security
Automation and secure access increase reliability and overall system efficiency.

Social Impact

Centralized Access
Patients and doctors benefit from easy access to complete medical records.

Emergency Safety
Improves safety and treatment quality, especially during emergencies.

Personal Health Awareness
Increases convenience and awareness of personal health information.

Economic Impact

Cost Reduction
Reduces costs by avoiding repeated tests and unnecessary paperwork.

Time Optimization
Saves valuable time for both patients and healthcare providers.

Resource Productivity
Improves productivity and supports efficient use of healthcare resources.

Challenges Faced

- **Medical Document Processing:** Extracting clinical information from different PDF formats accurately.
- **Data Preprocessing:** Handling missing, invalid, and inconsistent medical values.
- **ML Model Development:** Selecting and evaluating suitable models for risk prediction.
- **Explainable AI:** Making ML predictions understandable using SHAP explanations.
- **System Integration:** Connecting frontend, backend, database, document processing, and ML modules.
- **Data Security:** Implementing authentication, consent management, role-based access, and audit logging.
- **Clinical Safety:** Ensuring AI outputs support clinical decisions without presenting them as medical diagnosis.

Conclusion

The Medical Record Management System provides an effective solution to the problem of scattered healthcare data. By storing all medical records in a centralized digital platform, it helps doctors access complete patient information quickly and accurately. This reduces repeated tests, saves time, and improves the quality of treatment. It also ensures better coordination between different healthcare providers. The system supports continuous tracking of patient health over time. Overall, it creates a more organized and efficient healthcare environment.

The system ensures data security through authentication and role-based access, making it safe and reliable for all users. It also improves convenience for patients by allowing them to access their records anytime and anywhere. Overall, the project enhances healthcare efficiency, supports better decision-making, and provides a modern approach to managing medical records. It reduces dependency on paper-based records and minimizes data loss. The platform is user-friendly and easy to use for all types of users. It also lays a foundation for future improvements using advanced technologies like AI.

Future Work

- **Advanced ML Models:** Explore deep learning and advanced ensemble models for improved risk prediction.
- **Real-Time Monitoring:** Integrate wearable devices and real-time health data.
- **Mobile Application:** Develop Android/iOS apps for easier patient access.
- **Cloud Deployment:** Deploy the system on scalable cloud infrastructure.
- **Enhanced AI:** Add advanced LLM-based medical assistance and personalized health insights.
- **Interoperability:** Integrate with hospitals, laboratories, pharmacies, and standardized health-record systems.
- **Multilingual Support:** Provide healthcare information in multiple regional languages.

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THANK YOU